Solving systems of equations with the [redacted]ination method

1

a 7 b $x=-\frac{125}{17}$ c $y=\frac{39}{17}$

2

a $2x+3y=-7$ and $-9x+y=-4$ b $x=\frac{5}{29}$

3

a $-29x=6$ b $x=-\frac{6}{29}, y=\frac{48}{29}$

4 When we combine the equations, all the variables are eliminated and we end up with a false statement such as $0=2$.

5

a
- i Infinitely many solutions
- ii Dependent

b
- i No solutions
- ii Inconsistent

6 Multiply by 5

7

a Multiply the first equation by 7 and multiply the second equation by 6, and then subtract the equations.

b Multiply the first equation by 10 and multiply the second equation by 100 to remove the decimals. Then add the equations.

8

a $x=5, y=5$ b $x=-9, y=4$ c $x=6, y=7$ d $x=2, y=\frac{1}{5}$

e $x=-20, y=15$ f $x=\frac{1}{2}, y=\frac{3}{2}$ g $p=\frac{3}{5}, q=\frac{4}{5}$

9

a -1 b 59 c -11